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October 17, 2026

Reflection journal

For this week's module, we focused on regression and classification. I was mildly familiar with the topic going into it, due to my statistical programming class last semester. I am also taking a course on data mining as well, where we are also covering regression for this week's topic. So this lab acts as extra reinforcement for my knowledge of the subject. We start off with the titanic dataset for this assignment.

In task 1 of the lab, we train the model with linear regression to predict the fare of a passenger based on age and class. The findings:

- Coefficients(.coef_) : [Age: -0.43695889, Pclass: -38.29728935]
  This means that each year the age increases, the fare increases by $0.44. As the class changes (ie. 1st class -> 2nd class -> 3rd), the fare decreases as well.
- Mean square error for Fare prediction : 3364.92. The square root of this is 58.01, which means the model was off by about $58.01 on average.

In task 2, we use classification to predict rather a passenger survived based on their age, class, and sex. After training a model with an instance of Logistic Regression, the model output revealed an accuracy for survival prediction to be 74.83%

The main takeaways from this lab were learning how to utilize the regression and classification depending on what type of prediction I'm trying to achieve. I understood that regression works only for continuous numeric values, while classification is only effective to prediction output belonging to a category or label. I also learned the use of Mean Squared Error and accuracy_score; MSE is essential for regression because it measures the magnitude of an error between numbers, while accuracy_score measures the percentage of correct predictions.

Reflection – Lab 06: Regression and Classification

In this lab, I learned the difference between regression and classification models. Regression predicts numerical values, such as passenger fares, while classification predicts categories, such as whether a passenger survived. I used Linear Regression to predict fares and Logistic Regression to predict survival. I also learned how mean squared error and accuracy are used to evaluate different types of models.

One thing I learned was that the features used in a model can affect its predictions. Understanding the model's coefficients helped me see how Age and passenger class relate to predicted fares. Both models ran successfully, and I was able to review their results. My contribution was preparing the data, running the models, evaluating their results, and completing the knowledge check.

Elquin Ponce

Seth Alvarez

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October 17, 2026

Lab 06 Reflection: Regression and Classification

In this lab I built my first regression and classification models on the Titanic dataset. The code was the easy part in this lab, the harder task was trying to understand what the numbers represented. The coefficients were -0.44 for Age and -38.30 for Pclass. Each step down in class drops the predicted fare by about 38 dollars, so class matters a lot more than age. The Module 06 powerpoint slides say a bigger coefficient size means a stronger effect, and the results in this lab match that.

My biggest lesson from this lab came from the regression result. The MSE was 3364.92 and its square root was 58.01, so the model is off by about 58 dollars on average. This is clearly a big miss because many fares are smaller than that. Moreover, the Module 04 slides show that Fare has a long tail of high values, and the Module 06 slides say outliers pull the line, so with only two features the model can’t explain the high fares, so its weak even though the code is correct.

Overall, this lab showed me that a model that runs doesn’t mean that its the same as a model that performs well.

Reflection Journal & Knowledge Check- Diane Lenhoff

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Reflection – Linear Regression

In this exercise, I learned how Linear Regression uses different features to predict a numerical value. I used passenger age and class to predict ticket fares on the Titanic. My model had a Mean Squared Error of 3,364.92 and a Root Mean Squared Error of $58.01, which showed me that the predictions were not very precise.

I also found the coefficients interesting. Passenger class had a much larger effect on the predicted fare than age. This helped me understand how to interpret a model instead of just looking at its accuracy. If I were to improve the model, I would explore additional features to see whether they could reduce the prediction error.

Knowledge Check

1. In your own words, what is the key difference between a regression problem and a classification problem?

A regression problem predicts a numerical value, while a classification problem predicts a category or class. In this lab, we used regression to predict a passenger's fare and classification to predict whether a passenger survived or did not survive.

2. The LinearRegression model has an attribute called .coef_. After you train the model, print lr_model.coef_. What do these numbers represent?

The numbers are the coefficients for the features Age and Pclass. They show how much each feature affects the predicted fare. A positive coefficient means the predicted fare increases as that feature increases, while a negative coefficient means the predicted fare decreases, assuming the other feature stays the same.Predicted Fare = starting value + (Age × Age coefficient) + (Pclass × Pclass coefficient).

The coefficients show how Age and Pclass affect the predicted ticket fare. My Age coefficient was -0.437, meaning that each additional year of age decreases the predicted fare by approximately $0.44 when passenger class stays the same.

The Pclass coefficient was -38.30, meaning that moving from first class toward third class decreases the predicted fare by approximately $38.30 for each class level, assuming age stays the same.

Both coefficients are negative, showing that increasing either variable is associated with a lower predicted fare in this model.

3. Why did we use mean_squared_error to evaluate the regression model but accuracy_score for the classification model?** Why wouldn't accuracy be a good metric for the fare prediction task?

We use mean squared error for regression because it measures how far the predicted number is from the actual number. We use accuracy for classification because it measures how often the model predicts the correct category. Accuracy would not work well for fare prediction because the predicted fare does not need to exactly match the actual fare to be a good prediction.

KeiChara Fleeks-Jones

Lab 06 Reflection Journal

In this lab I learned the difference between regression and classification. Regression predicts continuous values while classification predicts categories such as if a passenger survived. I also learned that Logical Regression is indeed a classifier although it is not evident by the name. The Linear Regression model predicted fares with a Root Mean Squared Error of about $58, which shows that `Age` and `Pclass` alone aren't enough to reliably estimate ticket prices given how widely Titanic fares varied. The Logistic Regression model performed better, correctly predicting survival about 75% of the time, largely thanks to the strong predictive power of `Sex` and `Pclass`. Looking at the model coefficients helped me understand how these algorithms mathematically weigh each feature, especially how gender and passenger class shaped survival odds. Going forward, I'd like to add features such as family size.